Linux(Cohesity)

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Created by	Priyanshu K Sharma

🔷Linux Interview Q&A (Placement-focused)

1. What is Linux, and how is it different from Windows?

Linux is an **open-source operating system kernel** created by **Linus Torvalds** in 1991. It is commonly used in servers, embedded systems, networking devices, and also desktop environments. Distributions like Ubuntu, Fedora, RHEL provide a full OS built on the Linux kernel.

Windows is a **proprietary operating system** developed by Microsoft, widely used on desktops and enterprise systems.

Feature	Linux	Windows
Source Code	Open-source (GPL license)	Closed-source (proprietary)
Cost	Free (support may be paid)	Paid license required
Security	More secure (permissions, SELinux/AppArmor, open community audits)	More malware-prone due to popularity, patching needed
Customization	Highly customizable (kernel, shell, UI)	Limited customization
User Interface	CLI-focused (can install GUI)	GUI-first, CLI optional
Package Management	`apt`, `yum/dnf`, `zypper`	`.exe` or `.msi` installers
Usage	Servers, DevOps, Cloud, Networking	Personal PCs, Office environments

2. Difference between UNIX and Linux

Feature	UNIX	Linux
Origin	Developed in 1970s by AT&T Bell Labs	Developed in 1991 by Linus Torvalds
License	Proprietary (HP-UX, Solaris, AIX)	Open-source (GPL)
Portability	Hardware-dependent (specific vendors)	Portable, runs on almost all hardware
Cost	Paid, requires vendor support	Free (commercial support optional)
User Base	Large enterprises, legacy systems	Servers, desktops, cloud, mobile (Android)
Example Systems	IBM AIX, HP-UX, Oracle Solaris	Ubuntu, RHEL, CentOS, Fedora, Debian

👉 In short: Linux is a **free, open-source clone of UNIX** designed to be portable and community-driven.

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3. Difference between Linux distributions (RHEL, Ubuntu, CentOS, etc.)

Distribution	Type	Package Manager	Use Case
RHEL	Commercial, Enterprise	`yum/dnf` (RPM-based)	Enterprise servers, paid support
CentOS / Rocky / AlmaLinux	Community, Free RHEL clone	`yum/dnf` (RPM-based)	Servers, free alternative to RHEL
Ubuntu	Debian-based, user-friendly	`apt` (DEB-based)	Desktops, cloud servers, beginners
Debian	Stable, community-driven	`apt` (DEB-based)	Base for Ubuntu, servers
Fedora	Cutting-edge, upstream for RHEL	`dnf`	Developers, testing new features
SUSE / openSUSE	Enterprise and community	`zypper`	Enterprise systems, SAP workloads

👉 They all use the **Linux kernel**, but differ in **package management, release cycle, and support model**.

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4. Explain the Linux Boot Process

1. **BIOS/UEFI** → Initializes hardware, runs POST (Power-On Self Test).
2. **Bootloader (GRUB/LILO)** → Loads kernel into memory.
3. **Kernel Initialization** → Detects hardware, mounts root filesystem.
4. **Init/systemd** → First process (PID 1) starts system services.
5. **Runlevel/Target** → Defines the state (single-user, multi-user, GUI).
6. **Login Prompt** → User authentication begins.

👉 Command to check boot messages:

```
dmesg | less
```

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5. Runlevels (SysV) vs systemd Targets

Runlevel (SysV)	Equivalent systemd Target	Description
0	`poweroff.target`	Halt the system
1	`rescue.target`	Single-user mode
3	`multi-user.target`	Multi-user CLI
5	`graphical.target`	Multi-user + GUI
6	`reboot.target`	Reboot the system

👉 Check current target:

```
systemctl get-default
```

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6. Check Linux version and kernel version

- **OS version:**

  ```
  cat /etc/os-release
  lsb_release -a   # if available
  ```

- **Kernel version:**

```
uname -r
uname -a   # full details
```

👉 Example output:

```
Linux version 5.15.0-92-generic (Ubuntu 22.04)
```

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7. Difference between Absolute and Relative Paths

Path Type	Example	Description
Absolute Path	`/home/user/docs/file.txt`	Starts from root `/` , always the same location
Relative Path	`../docs/file.txt`	Relative to current working directory

👉 Use `pwd` (print working directory) to know where relative paths start.

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8. Difference between Soft Link and Hard Link

Feature	Hard Link	Soft Link (Symbolic Link)
Inode	Shares same inode as original file	Points to original file's path
Filesystem	Must be on same filesystem	Can span across filesystems
Deletion	File remains even if original is deleted	Breaks if original is deleted
Type	Acts like another copy of the file	Acts like a shortcut
Creation	`ln file link`	`ln -s file link`

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9. Check Disk Usage and Free Space

- **Filesystem usage (overall disk space):**

  ```
  df -h
  ```

  Shows free/used space for mounted filesystems.

- **Directory/File usage:**

```
du -sh /path
du -sh *   # disk usage of all items in current dir
```

👉 `-h` makes it human-readable (MB/GB).

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10. Check Memory and CPU Usage

- **Memory usage:**

  ```
  free -h
  ```

- **Processes and CPU load:**

  ```
  top
  htop   # (better UI, if installed)
  ```

- **CPU details:**

  ```
  lscpu
  ```

- **Per-process usage:**

  ```
  ps aux --sort=-%mem | head
  ps aux --sort=-%cpu | head
  ```

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11. Difference between Process and Thread in Linux

Feature	Process	Thread
Definition	Independent program in execution	Lightweight unit of a process
Memory	Has its own memory space (heap, stack, data segment)	Shares memory and resources of parent process
Context Switch	Expensive (needs full state save/restore)	Faster (less overhead)
Creation	`fork()` or `exec()`	`pthread_create()`

Feature	Process	Thread
Isolation	Processes are isolated	Threads are not isolated, risk of race conditions
Example	Running `firefox`	Each tab in Firefox as a thread

12. How to Check Running Processes

- `ps aux` → Snapshot of all processes.
- `top` → Real-time view of CPU/memory usage.
- `htop` → Interactive version of `top` (navigate with arrows).

13. Difference between `/etc/passwd` and `/etc/shadow`

File	Purpose	Security
`/etc/passwd`	Stores user account info (username, UID, GID, home directory, shell)	World-readable, password field contains `x` (placeholder)
`/etc/shadow`	Stores hashed passwords and aging info	Only root can read

14. Create a New User and Assign Password

```
sudo adduser john
sudo passwd john
```

- `adduser` → creates account, home directory.
- `passwd` → sets password.

15. Schedule a Job in Linux (cron vs at)

- **cron**: Repeated/scheduled tasks.

  Example: Run script every day at 5 AM →

  ```
  crontab -e
  0 5 * * * /home/user/script.sh
  ```

- **at**: One-time job.

  Example: Run script tomorrow at 6 PM →

  ```
  echo "/home/user/script.sh" | at 6pm tomorrow
  ```

16. File Permissions in Linux

- **r (read)** = 4 → view file contents.
- **w (write)** = 2 → modify/delete file.
- **x (execute)** = 1 → run file as program.
- Example: `rwxr-xr--`
  - Owner: read, write, execute
  - Group: read, execute
  - Others: read

17. Change Permissions and Ownership

- **Change permissions**:

  ```
  chmod 755 file.txt
  ```

- **Change owner**:

  ```
  chown user file.txt
  ```

- **Change group**:

  ```
  chgrp developers file.txt
  ```

18. Difference between `find` and `locate`

Command	Working	Use Case
`find`	Searches in real time by traversing directories	Accurate but slower

Command	Working	Use Case
`locate`	Uses prebuilt `mlocate` database	Faster but may not reflect latest changes

19. Difference between Daemon and Service

- **Daemon**: Background process without user interaction (e.g., `sshd`, `cron`).
- **Service**: Daemon that is managed by `systemd` or `init` (e.g., `nginx.service`).

20. Check Logs in Linux

- Logs are stored in **`/var/log/`**.
- Common logs:
  - `/var/log/syslog` → General system logs.
  - `/var/log/auth.log` → Authentication logs.
  - `/var/log/dmesg` → Kernel ring buffer.
- Commands:

```
tail -f /var/log/syslog
journalctl -xe
dmesg | less
```

🔷 Networking

1. Difference between TCP and UDP in Linux Networking

Feature	TCP	UDP
Type	Connection-oriented	Connectionless
Reliability	Ensures delivery with ACKs, retransmissions, error checking	No reliability, best effort
Speed	Slower (due to overhead)	Faster (minimal overhead)
Use Cases	Web (HTTP/HTTPS), Email (SMTP, IMAP, POP3), File Transfer (FTP,	Streaming (VoIP, Video), DNS, Online gaming

Feature	TCP	UDP
	SFTP)	
Example Command	`nc -l 8080` (TCP listener)	`nc -u -l 8080` (UDP listener)

2. Troubleshoot a Network Issue

Common Linux commands:

- `ping google.com` → Check connectivity.
- `netstat -tulnp` → Show active connections/ports.
- `ss -tulnp` → Modern alternative to netstat.
- `traceroute 8.8.8.8` → Show path to destination.
- `tcpdump -i eth0 port 80` → Capture packets on interface.

3. SELinux/AppArmor

- **SELinux (Security-Enhanced Linux)**: Mandatory Access Control (MAC) framework; policies define what processes can access. Modes: *enforcing*, *permissive*, *disabled*.
- **AppArmor**: Profile-based security (per application). Uses path-based rules instead of labels.
- **Difference**: SELinux = label-based, more complex; AppArmor = path/profile-based, easier.

4. Hard Disk Partitions: MBR vs GPT

Feature	MBR (Master Boot Record)	GPT (GUID Partition Table)
Partition Limit	4 primary partitions (or 3 primary + 1 extended)	Up to 128 partitions
Disk Size Support	Up to 2 TB	Up to 18 EB
Boot Support	Legacy BIOS	UEFI
Redundancy	Single boot record (fragile)	Multiple partition tables for redundancy
Usage	Older systems	Modern systems

5. LVM (Logical Volume Manager)

- LVM = flexible disk management, allows resizing and snapshots.

  Steps to create LVM:

```
pvcreate /dev/sdb
vgcreate vg_data /dev/sdb
lvcreate -L 10G -n lv_backup vg_data
mkfs.ext4 /dev/vg_data/lv_backup
mount /dev/vg_data/lv_backup /mnt/backup
```

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6. RAID (Redundant Array of Independent Disks)

RAID Level	Description	Pros	Cons
RAID 0	Striping	High speed	No redundancy
RAID 1	Mirroring	Redundancy	50% storage loss
RAID 5	Striping + Parity	Balance of speed + redundancy	Needs ≥3 disks, slower writes
RAID 10	Mirror + Stripe	High speed + redundancy	Needs ≥4 disks

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7. Difference: SAN, NAS, DAS

Storage Type	Description	Example
SAN (Storage Area Network)	Block-level storage over a network (FC/iSCSI)	Datacenter storage
NAS (Network Attached Storage)	File-level storage shared via NFS/SMB	Shared folders in office
DAS (Direct Attached Storage)	Directly connected to server (local HDD/SSD)	External HDD

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8. Mount an NFS Share in Linux

```
sudo apt install nfs-common -y
```

```
sudo mount -t nfs 192.168.1.100:/exports/data /mnt/data
```

- Permanent entry in `/etc/fstab`:

```
192.168.1.100:/exports/data  /mnt/data  nfs  defaults  0 0
```

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9. iSCSI (Internet Small Computer Systems Interface)

- Protocol to access block storage over IP network.

  Steps to configure:

```
sudo apt install open-iscsi -y
sudo iscsiadm -m discovery -t sendtargets -p 192.168.1.200
sudo iscsiadm -m node --login
lsblk  # New storage appears
```

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10. systemd vs init

Feature	init (SysVinit)	systemd
Startup Type	Sequential execution of scripts (`/etc/init.d/`)	Parallel startup with dependencies
Boot Speed	Slower	Faster
Service Management	Basic start/stop	`systemctl` with monitoring, restart policies
Logging	Syslog	Integrated journal (`journalctl`)
Usage	Legacy distros (RHEL 6, Debian 6)	Modern distros (RHEL 7+, Ubuntu 16+)

🔷 Troubleshooting Scenarios

1. A server is running slow, how do you troubleshoot?

- **Check CPU/Memory/Disk:** `top`, `htop`, `iostat`, `vmstat`.
- **Check Processes**: High CPU/memory usage → `ps aux --sort=-%cpu`.
- **Check Network:** `ping`, `traceroute`, `netstat`, `ss`.

- **Check Logs**: `/var/log/syslog`, `journalctl -xe`.
- **Check Disk Space**: `df -h`, `du -sh * | sort -h`.

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2. A user cannot log in — what steps will you take?

1. Check account status:

```
id username
grep username /etc/passwd
```

2. Verify shell: `/etc/passwd` → valid shell like `/bin/bash`.
3. Check password expiration: `chage -l username`.
4. Check `/etc/shadow` for locked account (`!!` or).
5. Look at authentication logs: `/var/log/auth.log`.

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3. Disk space is full, how will you find large files and free space?

- Check disk usage: `df -h`.
- Find top directories: `du -sh /* | sort -h`.
- Find large files:

```
find / -type f -size +500M -exec ls -lh {} \;
```

- Clean `/var/log`, `/tmp`, old backups.

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4. A service is not starting, how do you debug it?

1. Check status:

```
systemctl status service_name
```

2. View logs:

```
journalctl -u service_name --since "10 min ago"
```

3. Check config syntax: e.g., `nginx -t`, `apachectl configtest`.

4. Verify ports: `ss -tulnp | grep <port>`.

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5. Network is unreachable, how do you fix it?

- Check IP: `ip addr show`.
- Check routes: `ip route show`.
- Restart network service: `systemctl restart NetworkManager`.
- Test connectivity: `ping 8.8.8.8`, `ping google.com`.
- Check firewall: `iptables -L` or `ufw status`.

🔷 Theoretical Questions

1. What is Linux, and how is it different from UNIX?

- **Linux**: Open-source kernel (Linus Torvalds, 1991).
- **UNIX**: Proprietary OS family (AT&T, IBM AIX, HP-UX).
- **Key difference**: Linux = free, open-source; UNIX = commercial, closed source.

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2. Explain the Linux Directory Structure

- `/` → Root.
- `/bin` → Essential binaries.
- `/sbin` → System binaries.
- `/usr` → User programs/libraries.
- `/var` → Logs, spool, temp files.
- `/etc` → Config files.
- `/home` → User directories.
- `/tmp` → Temporary files.
- `/dev` → Device files.

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3. Relative vs Absolute Paths

- **Absolute path**: Starts from `/` (e.g., `/home/user/file.txt`).

- **Relative path**: Relative to current directory (e.g., `../file.txt`).

4. Hard Links vs Soft Links

- **Hard Link**: Points to same inode; file exists even if original is deleted.

  ```
  ln file1 file2
  ```

- **Soft Link (symlink)**: Points to path; breaks if target is deleted.

  ```
  ln -s file1 file2
  ```

5. Linux vs Windows File Systems

Feature	Linux	Windows
Common FS	ext4, xfs, btrfs	NTFS, FAT32, exFAT
Case Sensitivity	Case-sensitive	Not case-sensitive
Permissions	r, w, x per user/group/others	ACLs, inherited permissions
Path style	`/home/user`	`C:\Users\User`

6. File Permissions in Linux

- `r = 4` (read), `w = 2` (write), `x = 1` (execute).
- Example: `rwxr-xr--` → Owner full, group read+execute, others read only.

7. Difference between chmod, chown, and chgrp

- `chmod` → Change file permissions.
- `chown` → Change file owner.
- `chgrp` → Change group ownership.

8. Check System Uptime

```
uptime
who -b
cat /proc/uptime
```

9. Find Current Logged-in Users

```
who
w
users
```

10. Difference between /bin, /sbin, /usr/bin, /usr/sbin

Directory	Purpose
`/bin`	Essential user commands (ls, cp, mv).
`/sbin`	Essential system binaries (shutdown, ifconfig).
`/usr/bin`	Non-essential user commands (gcc, python).
`/usr/sbin`	Non-essential system binaries (apachectl, nginx).

11. Explain the Linux boot process.

1. **BIOS/UEFI** → Performs POST (Power-On Self Test).
2. **Bootloader (GRUB/LILO)** → Loads the kernel.
3. **Kernel Initialization** → Loads drivers, mounts root filesystem.
4. **init/systemd** → Starts services & targets.
5. **Login** → User authentication.

12. What are runlevels / systemd targets?

- **Runlevels (SysV init)**: Defines the system state.
  - `0`: Halt, `1`: Single-user, `3`: Multi-user (CLI), `5`: GUI, `6`: Reboot.
- **systemd Targets**: Modern replacement.
  - `poweroff.target`, `multi-user.target`, `graphical.target`.
- Command:
  - Check current → `runlevel` or `systemctl get-default`
  - Change → `systemctl isolate multi-user.target`

13. How do you check memory and CPU usage?

- **CPU**: `top`, `htop`, `mpstat`, `uptime`.
- **Memory**: `free -h`, `vmstat`, `cat /proc/meminfo`.
- **Windows Equivalent**: Task Manager, `perfmon`.

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14. What is the difference between a process and a thread?

- **Process**: Independent execution unit, has its own memory space.
- **Thread**: Lightweight unit within a process, shares memory/resources.
- Analogy: Process = Program, Threads = Tasks inside the program.

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15. Difference between ps, top, and htop.

- `ps`: Snapshot of processes (static).
- `top`: Real-time process monitoring.
- `htop`: Improved top with colors, scrolling, and interactive controls.

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16. How do you schedule tasks in Linux? Difference between cron and at.

- **cron**: Schedules repetitive tasks.
  - Edit → `crontab -e` → `0 2 * * * /backup.sh` (daily at 2 AM).
- **at**: Schedules a one-time job.
  - `at 14:00` → `./script.sh`

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17. Difference between /etc/passwd and /etc/shadow.

- `/etc/passwd`: Stores user account info (username, UID, GID, shell). World-readable.
- `/etc/shadow`: Stores encrypted passwords & aging policies. Root-only.

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18. How do you create a new user and assign groups?

```
# Create user
sudo useradd priyanshu
```

```
# Set password
sudo passwd priyanshu
# Add to group
sudo usermod -aG developers priyanshu
```

- Verify groups: `groups priyanshu`

19. Explain sticky bit, SUID, and SGID in Linux.

- **SUID (Set User ID)**: File runs with file owner's privileges. Example: `/usr/bin/passwd`.
- **SGID (Set Group ID)**: File runs with group's privileges; directories inherit group ownership.
- **Sticky Bit**: Only file owner can delete files in a shared directory (e.g., `/tmp`).

20. How do you check which processes are using a file or port?

- **File**: `lsof filename`, `fuser filename`.
- **Port**:
  - `lsof -i :80`
  - `ss -tulnp | grep 80`
  - `netstat -tulnp`

21. Difference between TCP and UDP

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (establishes a connection before data transfer)	Connectionless (sends data without setup)
Reliability	Reliable (acknowledgment, retransmission, ordering)	Unreliable (no guarantee of delivery/order)
Speed	Slower due to overhead	Faster due to minimal overhead
Use cases	Web (HTTP/HTTPS), email, file transfer	Streaming, VoIP, gaming, DNS queries

22. Troubleshooting network connectivity in Linux

Steps:

1. Check IP configuration: `ip addr show` or `ifconfig`
2. Test connectivity: `ping <host>`
3. Check route table: `ip route show` or `route -n`
4. Test DNS resolution: `nslookup <domain>` or `dig <domain>`
5. Check ports/services: `telnet <host> <port>` or `nc -zv <host> <port>`
6. View interface status: `ethtool <interface>`
7. Check firewall: `iptables -L` or `firewalld` / `ufw` status

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23. Difference between SAN, NAS, and DAS

Feature	DAS (Direct-Attached Storage)	NAS (Network-Attached Storage)	SAN (Storage Area Network)
Connection	Directly attached to server	Connected over network (Ethernet)	High-speed network (Fibre Channel/iSCSI)
Accessibility	Single server	Multiple clients via network	Multiple servers at block-level
Protocol	None (block-level)	File-level protocols (NFS, CIFS/SMB)	Block-level protocols (iSCSI, FC)
Use case	Small-scale, local storage	File sharing	Large-scale enterprise storage

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24. What is LVM? How to create and extend a logical volume

- **LVM (Logical Volume Manager)** allows flexible disk management by creating logical volumes on top of physical volumes.

Steps to create and extend LV:

1. Initialize physical volumes:

```
pvcreate /dev/sdb
```

2. Create a volume group:

```
vgcreate vg01 /dev/sdb
```

3. Create a logical volume:

```
lvcreate -L 10G -n lv01 vg01
```

4. Format and mount:

```
mkfs.ext4 /dev/vg01/lv01
mount /dev/vg01/lv01 /mnt
```

5. Extend logical volume:

```
lvextend -L +5G /dev/vg01/lv01
resize2fs /dev/vg01/lv01  # For ext4
```

---

25. Difference between ext3, ext4, and XFS

Feature	ext3	ext4	XFS
Max file size	2 TB	16 TB	8 EB
Max volume size	16 TB	1 EB	8 EB
Journaling	Yes	Yes (faster, extent-based)	Yes (efficient for large files)
Performance	Moderate	Better than ext3	Excellent for large files and parallel I/O
Backward compatible	Yes	Mostly	No

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26. What is RAID? Explain RAID levels

- **RAID (Redundant Array of Independent Disks)**: Combines multiple disks for redundancy or performance.

RAID Level	Description	Pros	Cons
0	Striping	High performance	No redundancy
1	Mirroring	Redundancy	50% storage efficiency
5	Striping + Parity	Balanced performance + redundancy	Min 3 disks, slower writes
6	Striping + Double Parity	Can tolerate 2 disk failures	Slower writes, min 4 disks

RAID Level	Description	Pros	Cons
10	RAID 1+0	High performance + redundancy	Needs even number of disks, costly

27. What is SELinux/AppArmor

- **SELinux** and **AppArmor** are Linux security modules that enforce **Mandatory Access Control (MAC)** policies.
- **SELinux** uses labels on files and processes, more granular and complex.
- **AppArmor** uses path-based profiles, easier to configure.

28. What is iSCSI, and where is it used

- **iSCSI (Internet Small Computer System Interface)** allows SCSI commands over TCP/IP networks to access remote block storage.
- **Use cases**: Connecting servers to SAN over standard networks without Fibre Channel hardware.

29. Difference between init and systemd

Feature	init (SysV)	systemd
Boot process	Sequential	Parallelized, faster boot
Service management	Scripts in /etc/init.d	`systemctl`, units, dependencies
Logging	Separate logs	Integrated journal
Popularity	Older	Modern Linux distributions

30. How to mount NFS and CIFS shares

- **NFS mount**:

```
mount -t nfs server:/export/path /mnt
```

- **CIFS/SMB mount**:

```
mount -t cifs //server/share /mnt -o username=user,password=pass
```

31. A user cannot log in — troubleshooting

- Check `/etc/passwd` and `/etc/shadow` entries.

- Verify account status: `passwd -S <user>` or `chage -l <user>`
- Check shell: `grep <user> /etc/passwd`
- Check home directory permissions
- Inspect authentication logs: `/var/log/auth.log` or `/var/log/secure`

32. Server is running slow — steps to take

- Check CPU/memory usage: `top`, `htop`, `vmstat`
- Check disk I/O: `iostat -x`, `iotop`
- Check running processes: `ps aux --sort=-%cpu`
- Check network load: `iftop`, `netstat -i`
- Check logs for errors

33. Disk space full — how to find and clear

- Check usage: `df -h`
- Find large files: `du -sh /*` or `find / -type f -size +100M`
- Clear logs or temp files: `/var/log`, `/tmp`
- Clean package cache: `yum clean all` or `apt clean`

34. Service not starting — how to debug

- Check status: `systemctl status <service>`
- Start manually and check logs: `journalctl -u <service>`
- Check config files for errors
- Check dependencies: `systemctl list-dependencies <service>`

35. Network unreachable — troubleshooting

- Check interface status: `ip a`, `nmcli`
- Ping gateway, DNS, external IP
- Check routes: `ip route`
- Check firewall/iptables: `iptables -L`
- Check logs: `dmesg | grep eth`

36. Server not booting — how to fix

- Check bootloader: GRUB configuration
- Boot into single-user mode or rescue mode
- Check disk and filesystem: `fsck`
- Check recent kernel updates or config changes

37. Logs filling up disk — handling

- Rotate logs: `logrotate`
- Archive old logs: `tar czf /backup/logs.tar.gz /var/log/*.log`
- Delete unnecessary logs
- Configure systemd journal: `SystemMaxUse`, `SystemKeepFree`

38. Suspected high I/O wait — confirm and fix

- Check I/O stats: `iostat -x 1`, `vmstat 1`
- Check processes causing I/O: `iotop`
- Fix: Optimize disk usage, move high I/O processes, add faster storage (SSD), RAID optimization

39. Recover deleted file in Linux

- If using ext3/ext4, tools like `extundelete` or `testdisk`
- If using LVM snapshots, restore from snapshot
- If backup exists, restore from backup

40. Secure SSH on a Linux server

- Change default port from 22
- Disable root login: `PermitRootLogin no`
- Use key-based authentication, disable password auth
- Enable fail2ban or firewall rules
- Limit users allowed: `AllowUsers user1 user2`
- Keep SSH updated